

Modeling Continuous-time Interval Graphs using Marked Hawkes Process Network Models

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Abstract. In many application settings involving dynamic networks, the edges denote quantities that have durations, such as lengths of phone calls between two people. Such temporal networks with edge durations have often been referred to as *interval graphs*, as edges are active over time intervals. Most existing continuous-time network models are based on relational event models, which only model instantaneous events, ignoring their duration. In this paper, we propose a general approach for modeling continuous-time interval graphs by using marked Hawkes processes to jointly model edge start times (using the Hawkes process event times) and edge durations (using the marks). We use our modeling approach to extend the recently proposed Latent Space Hawkes (LSH) model, to continuous-time networks with edge durations. We demonstrate that our model achieves competitive prediction accuracy for both edge start times and durations while offering strong interpretability, yielding meaningful insights when used for exploratory analysis on a bike sharing network.

Keywords: Temporal network · Dynamic network · Continuous-time network · Edge duration · Relational event model · Temporal point process · Latent variable model.

1 Introduction

Many complex systems, such as social interactions, communication networks, and transportation systems, can be modeled by networks that vary continuously over time through a sequence of relational events [1]. Each relational event is commonly represented as a triplet (u, v, t) , indicating an instantaneous interaction between node u and node v at time t . A sequence of such instantaneous interactions is also known as a contact sequence [4]. Some types of interactions are not instantaneous and instead have durations, such as a phone call between two people (as compared to a text message, which is instantaneous). Such interactions with durations can also be modeled by relational events, where the triplet notation (u, v, t) is extended to a quadruplet (u, v, t, m) , with m denoting the duration of the event.

A continuous-time network constructed of interactions with durations is also referred to in the literature as an *interval graph* [4]. The toy example in Fig. 1

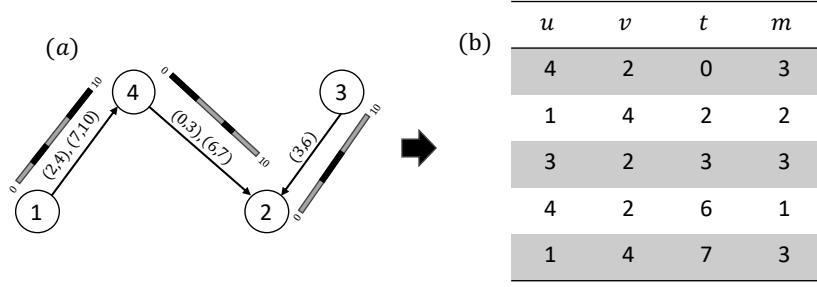


Fig. 1. Toy example of an interval graph (a) and its corresponding relational event table (b). In the interval graph, the overall time windows are visualized by the timeline bar, ranging from 0 to 10. Times where two nodes have an active edge are filled with black; the remaining times are filled with gray. The edge start and end times are also marked on the edges of the interval graph. In the relational event table, each row indicate an temporal edge from node u to node v at time t , with a duration m .

shows an interval graph with four nodes and five edges with durations over different time periods. As shown, the interval graph can be represented by a relational event table where each edge between a pair of nodes is represented by a quadruplet (u, v, t, m) .

There have been significant developments in relational event models (REMs) since their introduction by Butts [1]. REMs have been used extensively in the literature to model continuous-time networks in the form of contact sequences, i.e., instantaneous edges. There has been comparatively less literature on modeling interval graphs, i.e., continuous-time networks where interactions have durations.

In this paper, we introduce an approach for modeling continuous-time interval graphs by building upon REMs. We assume that each event is associated with a timestamp, which we use to model the edge start time, as well as a *mark*, which we use to model the edge duration. We introduce a new modeling approach for continuous-time interval graphs using marked Hawkes process models [9] with latent structures on the nodes. We use this modeling approach to develop interval graph versions of the recently proposed Latent Space Hawkes (LSH) [5] model. We demonstrate that models with different latent structures for the start times and durations of edges (dual latent structures) often yield improved prediction accuracy compared to models with shared latent structure while preserving strong interpretability.

2 Background

2.1 Latent Space Hawkes Model

Huang et al. [5] proposed the Latent Space Hawkes Model (LSH), which incorporates the latent space assumption originally proposed by Hoff et al. [3] into a mutually exciting Hawkes process [6] to model continuous-time networks

with instantaneous events. Each node u is mapped into a low-dimensional latent vector $\mathbf{z}_u$. The baseline rate μ_{uv} of the Hawkes process modeling edges from node u to v depends on their distances in this latent space, so that two nodes that are closer in the latent space have a higher chance of forming edges: $\mu_{uv} = e^{-\theta_1 \|\mathbf{z}_u - \mathbf{z}_v\|_2^2 + \theta_2 + \delta_u + \gamma_v}$, where θ_1, θ_2 are slope and intercept terms used to model the relationship between the squared distances between the latent positions of the nodes and their baseline rate. μ_{uv} is also affected by node-specific parameters δ_u and γ_v , denoting the sender and receiver propensities, respectively.

The conditional intensity function (CIF) of the LSH model is

$$\lambda_{uv}(t|\mathcal{H}_t) = \mu_{uv} + \sum_{t_{uv} < t} \sum_{q=1}^Q c_q \alpha_1 \beta_q e^{-\beta_q(t-t_{uv})} + \sum_{t_{vu} < t} \sum_{q=1}^Q c_q \alpha_2 \beta_q e^{-\beta_q(t-t_{vu})}$$

for all $u \neq v$. The parameters α_1, α_2 denote the jump sizes for self and reciprocal excitation in the LSH model. A sum of Q exponential kernels is used, with a set of fixed known decay rates $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_Q)$ and a set of scaling parameters $\mathbf{c} = (c_1, c_2, \dots, c_Q)$ for the kernels that sum to 1.

2.2 Related Work

One approach for modeling interval graphs is to assume that each edge switches from active to inactive and vice versa following a continuous-time Markov process. Zhang et al. [14] utilize this approach, assuming that all pairs of nodes without an active edge form one with rate λ , and all active edges get de-activated with rate μ . However, they assume that the network is only observed at discrete time snapshots, so that the interval graph is not actually continuous time. On the other hand, Rastelli and Fop [11] use two exponential distributions with different parameters to model inter-event times and durations, so that the interval graph is observed in continuous time. A significant limitation of [11] is the assumption that interaction and non-interaction lengths are captured by homogeneous Poisson processes. In particular, edge start times in many application settings are better modeled by non-homogeneous TPPs, such as multivariate Hawkes process, which are self and mutually exciting.

3 Proposed Model

3.1 General Modeling Approach

As shown in Fig. 1, we represent interval graphs using a sequence of marked timestamped events $E = \{(u_1, v_1, t_1, m_1), (u_2, v_2, t_2, m_2), \dots, (u_L, v_L, t_L, m_L)\}$, where L denotes the number of temporal edges, and the marks denote edge durations. A natural approach to model marked timestamped events is to use a marked temporal point process (TPP), such as a marked Hawkes process [9].

To model an interval graph, we augment a continuous-time network model with a model for the marks, i.e., the edge durations. We assume that marks

have no impact on the CIF of events and that the distribution of a mark m at timestamp t is independent of previous sequence of timestamps and marks (t_l, m_l) , where $t_l < t$. Our approach can build upon any TPP-based network model for edge start times, including continuous-time latent space models [5, 13].

We assume a latent variable representation for the marks, as durations of edges between different nodes may follow different distributions. For example, a person’s call durations may differ significantly when they are calling a coworker compared to their spouse. Our proposed modeling approach can be applied with any probability distribution to model the marks. Since marks correspond to edge durations, we use the log-normal distribution $\text{lognorm}(\eta, \sigma^2)$, parameterized by the log-scale mean η and log-scale variance σ^2 , which has been found to provide good fits to phone call durations [8]. We assume that the parameters for these distributions are specific to each node pair (u, v) and depend on the latent variables for the nodes u and v .

As shown in Fig. 2, we consider two variations of the latent representations for edge start times and durations. The shared representation ties together a node’s latent variables for generating both timestamps and marks so that both can be used to estimate the latent representations. The dual representation is inspired by the dual latent space model [13], which has separate latent spaces to model homophily and reciprocity. In our setting, having separate representations for timestamps and marks allows for additional flexibility at the cost of potentially higher variance estimates.

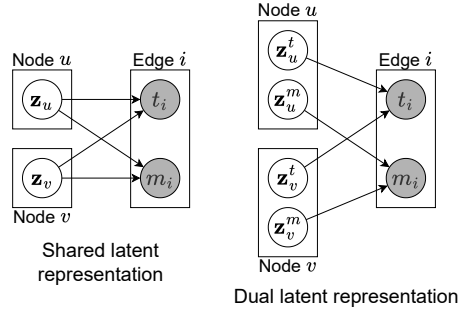


Fig. 2. Comparison of shared and dual latent representations for timestamps (edge start times) and marks (edge durations). Observed variables are shaded, while latent variables are not.

3.2 Latent Space Hawkes Interval Model

To extend the LSH model introduced in Section 2.1 to continuous-time networks with edge durations, we model events between each node pair by a marked bivariate Hawkes process. To integrate marks into this model, we consider two variants where event timestamps and marks arise from either dual latent spaces (LSH-Interval-D) or a shared latent space (LSH-Interval-S).

Each node u in the original LSH model has a latent d -dimensional vector z_u corresponding to its node position in the latent space. In the LSH-Interval-D model, each node has two latent vectors z_u^t, z_u^m corresponding to their node positions in the timestamp and mark latent spaces. These latent node positions are sampled from zero-mean spherical Gaussian distributions parameterized by the variances $\sigma_{z_t}^2$ and $\sigma_{z_m}^2$ for the timestamp and mark latent positions, respectively. To obtain the LSH-Interval-S model, consider a single shared latent space so that $z_u = z_u^t = z_u^m$.

The timestamps of events between node pairs (u, v) and (v, u) in the LSH-Interval-D model follow a bivariate Hawkes process, which generate the edge start times. All node pairs share the common parameter set $\Theta^t = (\theta_1, \theta_2, \alpha_1, \alpha_2, \mathbf{c}, \boldsymbol{\delta}, \boldsymbol{\gamma})$. The mean of the mark distribution for a node pair (u, v) , corresponding to the mean of the edge duration for (u, v) , follows a linear model of the squared distance between the latent positions of the nodes in the mark latent space, which we denote by $\mathbf{z}_u^m, \mathbf{z}_v^m$. For a log-normal distribution, we model the log-scale mean using $\eta_{uv} = -\theta_3 \|\mathbf{z}_u^m - \mathbf{z}_v^m\|_2^2 + \theta_4$, where θ_3, θ_4 denote the slope and intercept parameters, respectively. We model the log-scale variance using a fixed constant σ_m^2 for all node pairs. All node pairs share a common set of parameters for marks denoted by $\Theta^m = (\theta_3, \theta_4, \sigma_m^2)$.

To fit the model, we follow the approach used in the original LSH model outlined in [5], we apply the L-BFGS-B algorithm to minimize the negative log-likelihood for both event timestamps and marks. We initialize the latent positions for both event and mark latent spaces using multidimensional scaling, while all other parameters are set to random initial values.

4 Experiments

We evaluate our proposed LSH-Interval models on 4 real interval graph datasets: MIT Reality Mining [2], Militarized Interstate Disputes (MID) [10], High School Student Interaction [7], and London Bike Share [12]. We compare both versions of our model (single and dual latent structures) with two existing approaches. First is LSH + log-normal, where we fit an LSH model only to the edge start times and then fit a single log-normal for the marks (edge durations). We also compare to the ExpSBM model proposed by Rastelli and Fop [11].

4.1 Prediction Accuracy

To evaluate prediction accuracy, we split each dataset: 80% for training and the remaining 20% for testing. Model parameters are estimated on the training set, and prediction accuracy is evaluated on the test set. The number of latent dimensions d are selected to maximize the log-likelihood on the test set.

We evaluate the predictive ability of our interval graph models using predictions of future events and durations. We randomly sample 1,000 time points within the test set period. For each time point t , we locate the next actual event that occurs after t along with its corresponding duration and the associated node pair. Given this node pair, we perform both event timestamp and mark prediction. This process allows us to forecast the duration of future interactions between nodes. To evaluate the accuracy of our predictions, we compute the Root Mean Square Error (RMSE) between the predicted and actual quantities.

The future event and mark prediction results are shown in Tables 1 and 2, respectively. For event timestamp prediction, our proposed interval models consistently achieve better results than ExpSBM. Among all our proposed interval models, LSH-interval-D achieves the best future event prediction for all

Table 1. Evaluation of RMSE for future edge start time (event timestamp) prediction for different models. The lowest RMSE for each dataset is shown in bold.

Model	Reality	MID	Bike Share	High School
LSH-interval-S	29.9	30.6	52.3	64.0
LSH-interval-D	29.9	29.3	52.0	53.2
LSH + log-normal	30.9	29.8	54.8	56.5
ExpSBM	48.8	34.5	92.9	129

Table 2. Evaluation of RMSE for future edge duration (mark) prediction for different models. The lowest RMSE for each dataset is shown in bold.

Model	Reality	MID	Bike Share	High School
LSH-interval-S	0.0124	8.45	1.48	1.06
LSH-interval-D	0.0127	6.57	1.51	0.997
LSH + log-normal	0.0126	8.74	1.50	1.04
ExpSBM	0.0113	4.37	1.52	1.41

datasets. For mark prediction, our proposed model outperforms ExpSBM on the Bike Share and High School datasets but performs worse on the Reality and MID datasets. The differences in mark prediction are likely attributed to the different assumptions in distribution for the edge durations: log-normal for the LSH methods compared to exponential for the ExpSBM. On the other hand, the differences in event timestamp prediction are likely due the differences in the underlying TPP models: mutually exciting Hawkes processes for the LSH methods compared to homogeneous Poisson processes for ExpSBM.

4.2 Model-based Exploratory Analysis

We now present a case study demonstrating the application of our proposed LSH-Interval models for exploratory analysis on the London Bike Share data. In this network, timestamped edges represent instances where a person rents a bike from a start bike station (start node) and returns it to an end bike station (end node). The duration of the rental is used as the mark for each edge.

We use the *negative slope* model as discussed in [5] by forcing $\theta_3 < 0$. In this case, bike stations with longer rental times tend to be placed farther apart in the mark latent space. Since bike rental time heavily depends on the physical distance between stations, those farther apart on the real-world map are likely to have longer rental times.

In Fig. 3, we present the 2-D mark latent space plot fitted to the Bike Share data. The top longest average bike rental times between pairs of stations are highlighted with red lines, while the top shortest average bike rental times are marked with blue lines. Notice that node pairs with longer marks tend to be placed far apart in the mark latent space. On the other hand, node pairs with shorter average marks tend to be placed closer together in the mark latent space.

We also mapped the pairs of bike stations highlighted in Fig. 3 onto a physical map, using OpenStreetMap, as shown in Fig. 4. In practice, when two stations

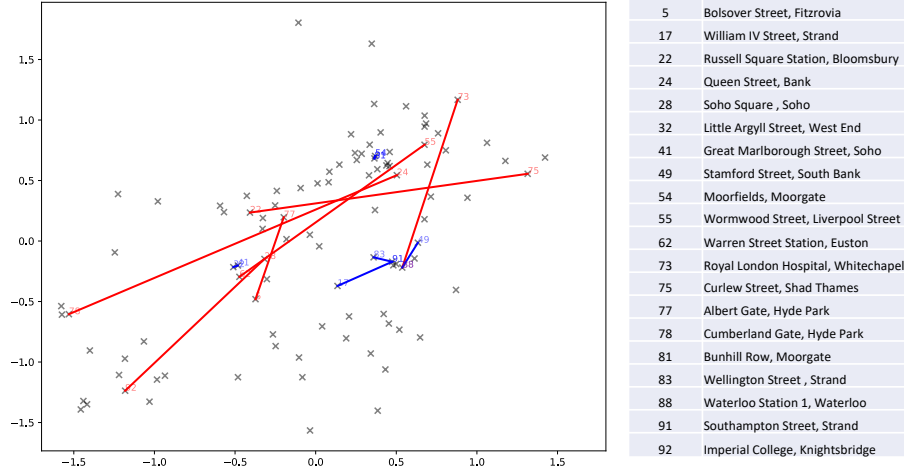


Fig. 3. 2-D mark latent space for LSH-Interval-D fit to Bike Share data. Edges highlighted with red (blue) denote the longest (shortest) average connections between pairs of bike stations in London. Names of highlighted bike stations are listed on the right.

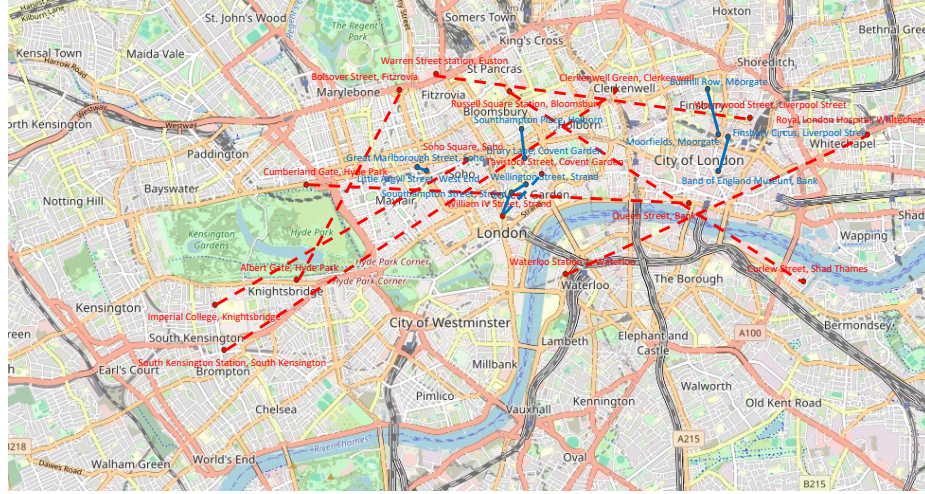


Fig. 4. Physical map for Bike Share data. The pairs of stations shown in Fig. 3 are marked on this physical map. Similarly, pairs of stations with the longest average connections are highlighted with red lines, while pairs with the shortest average connections are marked with blue lines.

are physically far apart, bike rentals between these stations tend to have longer durations because users need more time to return the bike over a longer distance. Comparing Fig. 3 and 4, our mark latent space plot makes sense: stations that are physically far apart tend to have longer rental times and are thus more likely to be placed far apart in the mark latent space, as it requires more times for returning bikes over longer distances.

This case study demonstrates the utility of our proposed LSH-Interval-D model for exploratory analysis, as the distances between latent positions of the bike stations match up well with their physical distances. We believe that this type of model can be useful in many other settings involving interval graph data.

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